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A Practical Guide to Machine Learning in Petrophysics: Navigating the Chaos

S. K. Sarkar, S. Dasgupta, A. Chapellin Biundo, and O. Berkimbay, SLB, Doha, Qatar

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Abstract

The integration of machine-learning (ML) into petrophysical workflows has transformed formation evaluation and reservoir characterization. However, the overwhelming variety of ML approaches (algorithms/ workflows) and data challenges often hinder effective implementation. This paper provides a structured guide to applying ML in petrophysical analysis, covering data preprocessing, outlier detection, algorithm selection and model interpretation, keeping practical considerations in mind.

The study follows a systematic approach to implementing ML in petrophysics. First, petrophysical data types (wireline, Logging While Drilling (LWD), core) are reviewed, along with pre-processing techniques such as depth matching, normalization, and missing data handling. Next, outlier detection methods (statistical, ML-based) to enhance data reliability are explained and discussed. Various ML algorithms—including linear regression, decision trees, random forests, support vector machines, and neural networks—are evaluated for their suitability in formation evaluation. This will be done using multiple case studies to demonstrate the entire workflow from data preparation to model validation, offering practical insights.

This review paper will reveal key insights into the strengths and limitations of ML models in petrophysics. Feature selection and engineering significantly impact model accuracy, with decision trees and random forests outperforming traditional regression techniques in capturing nonlinear petrophysical relationships. Neural networks show promise in handling complex interactions but require careful hyperparameter tuning and large datasets.

Outlier detection techniques successfully improve data reliability, reducing errors in ML model predictions. The results emphasize the importance of interpretable ML models to ensure that petrophysical insights remain geologically consistent. While ML models enhance predictive capabilities, they must be used in conjunction with domain expertise and traditional petrophysical interpretation to avoid misleading results.

The findings demonstrate that a structured ML workflow—starting with quality data preprocessing and outlier detection—leads to more robust and reliable predictions. However, challenges remain, including data availability, small sample sizes, and the need for interpretability in ML models used for decision-making.

This paper provides a practical roadmap for applying ML in petrophysical workflows, addressing common data challenges and algorithm selection pitfalls. Unlike existing studies, it offers a step-by-step

implementation strategy, highlighting real-world challenges and solutions in ML adoption for formation evaluation. The structured framework will serve as a valuable reference for petrophysicists seeking to integrate ML into their reservoir characterization workflows effectively.

Introduction

The conventional approach in the oil and gas industry has been to apply different correlation models and that is completely true for the petrophysics discipline as well, but the increasing advent of machine-learning (ML) has helped automate petrophysical interpretations and improve predictions, however one needs to carefully integrate ML with proper domain knowledge to ensure robust and meaningful insights.

The biggest advantage of ML is revealed by integration with traditional petrophysical methods, which would help us make more informed decisions about reservoir development. ML models operate on numerical data and are grounded in mathematical rigor – offering consistent, data-driven outputs. However, numerical data can be misinterpreted without geological and petrophysical context and that is where the domain expert needs to step in.

The number of publications related to ML in petrophysics has been steadily increasing, with a particularly notable surge observed from 2019 onward. Estimates suggest that publication activity grew by approximately 30% during the years 2023 and 2024 alone. While it remains challenging to quantify the exact number of research papers published between 2020 and 2024—due to the interdisciplinary scope and broad application of ML in geosciences—several bibliometric analyses and systematic reviews have identified consistent upward trends in both volume and diversity of ML applications within petrophysical domains. Now, one of the biggest issues of this dizzying number of publications is shifting out the noise from the signal. There is an overwhelming amount of content and most of the publications are location-specific/ reservoir-specific, so the entire purpose of this publication is to try to consolidate the information from multiple sources and attempt to provide some guidance, so as to implement ML in petrophysical studies with a structured workflow, as will be discussed in upcoming sections.

Benefits of ML in Petrophysics

The integration of ML into petrophysical workflows presents a transformative shift in how subsurface data is interpreted and utilized. Petrophysical data—primarily obtained from downhole logging tools and core analysis—are generally well-structured, physically modeled, and depth-indexed. These characteristics make them highly amenable to ML implementation, where algorithms can be trained and validated with confidence. Despite the challenges introduced by varying data scales, types, and dimensionalities, ML excels at uncovering patterns and nonlinear relationships within complex datasets, facilitating robust and scalable predictive solutions (Xu et al., 2022).

ML proves particularly valuable when physics-based models are unavailable or impractical due to complexity, computational cost, or time constraints. Even in the presence of data uncertainty or incompleteness, ML enables actionable decision-making, bridging the gap where conventional methods fall short (Ma and Amabeoku, 2015). This capability empowers petrophysicists to generate reliable outputs that support business strategies under operational time pressure.

Another critical advantage of ML lies in its ability to automate repetitive and labor-intensive tasks. Processes such as data cleaning, labeling, clustering, and well log correlation can be streamlined through automation, leading to enhanced productivity and consistency. Akkurt et al. (2018) reported that ML models replicated expert interpretations with over 85% accuracy across 59 wells, dramatically accelerating project timelines and enabling rapid decision support. Furthermore, ML's mathematical objectivity reduces human-induced biases and promotes reproducibility (Brazell et al., 2019).

ML also offers unique advantages in analyzing high-dimensional data, such as images, arrays, and 3D CT volumes, which typically exceed human interpretive capacity. ML models can extract complex spatial and

textural features at a scale and resolution unattainable through manual interpretation, enabling consistent classification and unlocking new insights from underutilized datasets (Xu et al., 2022).

Limitations of ML in Petrophysics

Despite its growing utility in petrophysical analysis, machine learning (ML) is not without limitations. One of the most critical challenges lies in data correctness and representativeness. Petrophysical data may be accurate in measurement but unrepresentative due to factors such as tool calibration issues, sampling errors, or physical alterations during core handling. ML models trained on such data risk generating misleading predictions, as the algorithms lack the capacity to recognize non-representative inputs without expert intervention (Ma and Amabeoku, 2015; Akkurt et al., 2018).

Data consistency further complicates ML workflows. Petrophysical datasets often originate from different wells, fields, or service providers, leading to inconsistencies in acquisition protocols, tool generations, and geological conditions. These variations undermine comparability and introduce noise that can obscure true relationships. Normalization and correction are essential but technically demanding steps (Xu et al., 2022).

While ML thrives on large datasets, data size, quality, and relevance remain limiting factors in petrophysics. High-resolution core data are typically sparse and attempts to augment datasets with information from disparate fields may degrade model performance due to geological non-analogues and inconsistent measurement practices. In many cases, a smaller but more relevant dataset yields better model accuracy (Xu et al., 2022).

Another concern is the accuracy of data labeling, particularly for supervised learning. Errors in core measurements or biases in geological facies labeling propagate directly into ML models, compromising their reliability. ML cannot independently assess label quality, making rigorous data validation imperative (Xu et al., 2022).

Generalization is another inherent limitation. Most ML models perform well only within the domain of their training data, often failing when applied to different formations or geological contexts (Xu et al., 2020).

Finally, as statistical learners, ML models tend to average outliers and struggle with predicting extreme values. The 2021 SPWLA ML contest highlighted this limitation, with all teams overestimating shear slowness in high-velocity zones (Yu et al., 2021).

Petrophysical Data in Machine Learning

Petrophysical data form the foundation of reservoir characterization, capturing the physical and chemical properties of subsurface formations. These data are derived from various sources such as well logs, core analyses, mud logs, seismic attributes, and production records. In the context of machine learning (ML), such datasets act as both features (inputs) and labels (targets), enabling predictive, diagnostic, and prescriptive modeling of reservoir properties and behavior. A robust understanding of these data types, their formats, and resolution mismatches is essential for meaningful ML implementation, preventing models from becoming black-box approximations devoid of geological credibility.

Petrophysical Data Types and Resolution Considerations

Petrophysical workflows incorporate a diverse set of data sources:

Well Logs. Acquired through wireline and LWD tools, these provide high-resolution (centimeter scale) continuous depth-based measurements such as gamma ray, resistivity, density, and sonic velocity.

Core Data. Discrete measurements (sparse intervals and limited depths) but precise laboratory measurements (e.g., porosity, permeability, capillary pressure) that serve as ground truth references.

Mud Logs. Qualitative to semi-quantitative records obtained in real-time during drilling. They include lithological descriptions, hydrocarbon shows, and gas readings at a coarser resolution.

Seismic Attributes. Provide extensive areal coverage but low vertical resolution. Attributes like acoustic impedance and amplitude inform structure and stratigraphy.

Production and Pressure Data. Time-series measurements that track fluid flow, pressure variations, and operational metrics throughout the field life.

The heterogeneity across these datasets—in resolution, sampling interval, and spatial scale—poses a major challenge in ML applications. For instance, seismic data represents volumes in the range of tens of meters, while well logs capture changes at the centimeter scale, and core data are restricted to selected depth intervals. Without proper scale harmonization, ML models may create a false relationship and show biasness towards dominant patterns in coarse data (e.g., seismic) while overlooking high-resolution heterogeneity captured in well logs and cores. As demonstrated by [Zhang et al. \(2019\)](#), such inconsistencies can mislead models into generating geologically implausible outputs.

To mitigate this, alignment strategies are critical. For example, in reservoir property modeling seismic data should be resampled to match the vertical resolution of logs. On the other hand, in seismic-scale models logs and core data may need upscaling. [Figure 1](#) presents a conceptual workflow that guides the integration of multiscale data in a machine learning context. [Table 1](#) in this section summarizes preprocessing techniques aligned with data types and suitable ML algorithms. Additionally, advanced ML architectures like convolutional neural networks (CNNs) with skip connections have shown success in learning multiscale features, preserving both resolution fidelity and geologic meaning ([Gul et al., 2020](#)). [Zhang et al. \(2019\)](#) and [Alaudah et al. \(2019\)](#) successfully demonstrated the Seismic-Log-Core integration using CNNs in shale reservoir characterization study.

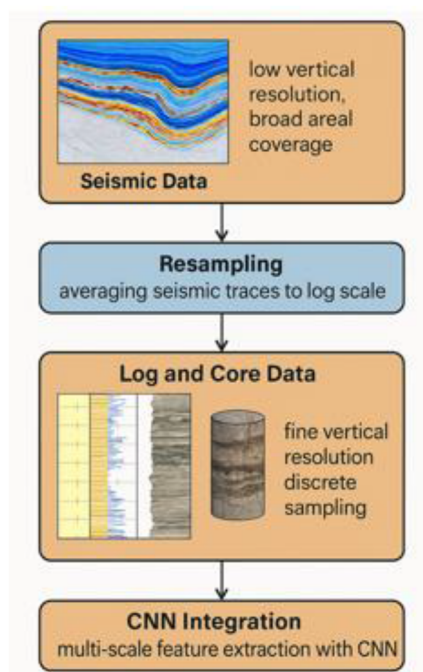


Figure 1—Workflow for Scale Alignment and Data Integration Across Seismic, Log, and Core Domains.

Table 1—Petrophysical Data Types, Preprocessing Techniques, and Compatible Machine Learning Algorithms:
This table provides a structured overview of common petrophysical data types—ranging from continuous numeric logs to unstructured textual and time-series data—along with recommended preprocessing methods, suitable machine learning algorithms, and typical use cases. It serves as a practical guide for petrophysicists to align data characteristics with appropriate ML workflows, ensuring geologically meaningful and technically robust outcomes.

Data Type	Examples	Preprocessing Techniques	Compatible ML Algorithms	Use Cases
Continuous Numeric	Gamma Ray (GR) Resistivity Porosity Neutron/Density	Z-score or Min-Max normalization; Missing value imputation; Smoothing filters; Depth alignment	Linear/Multiple Regression, Random Forest Regressor, XGBoost, Artificial Neural Networks (ANN)	Predicting porosity, water saturation, permeability (Zhang et al., 2019; Mohaghegh, 2017)
Categorical/Discrete	Lithofacies Well Status Rock Type	One-hot encoding; Ordinal encoding; Handling missing categories	Decision Trees, Random Forest Classifier, Support Vector Machines (SVM), Logistic Regression	Facies classification, lithology prediction (Gul et al., 2020; Alaudah et al., 2019)
Image-Based	Borehole Images Core photos	Rescaling; Noise filtering; Edge detection; Image augmentation	Convolutional Neural Networks (CNNs). Transfer Learning (ResNet VGG), Autoencoders	Fracture classification, textural facies detection, core analysis (Alaudah et al., 2019; Karimpouli et al, 2020)
Textual/Unstructured	Mud Logs. Litho-descriptions	Text cleaning; Stop-word removal; Tokenization; Vectorization (TF-IDF, Word2Vec, BERT]	Natural Language Processing (NLP) Classifiers, Recurrent Neural Networks (RNNs), BERT	Extracting lithology, identifying formation tops, event detection (Alaudah et al., 2019)
Time-Series	Production Pressure Injection Rates	Time alignment; Lag feature generation; Rolling statistics	RNNs, LSTMs, Temporal CNNs, Hybrid ARIMA-ML models	Production forecasting, pressure anomaly detection (Tachmazidis et al., 2022; Mohaghegh, 2017)

Table 2—Here is a comparison of some of the major (not fully comprehensive) outlier detection techniques.

Method	Type	Strengths	Limitations	Best Use Case
Z-score/Modified Z-score	Statistical	Simple, fast, effective on normally distributed data	Assumes normality, less effective on skewed or data	Basic outlier flagging on single logs
Interquartile Range (IQR)	Statistical	Robust to outliers, non-parametric	May miss context-dependent anomalies	Depth-based logs (e.g., GR, TNPH)
Boxplots	Visual	Easy to interpret. quick identification	Not scalable for large datasets	QC of a few wells or log curves
Cross-plots	Visual	Reveals physical inconsistencies between logs	Requires domain expertise to interpret	Comparing porosity, resistivity, density, etc.
Log-Depth Plots	Visual	Captures vertical anomalies and tool spikes	Manual, subjective	Spotting logging tool failures or borehole issues
Isolation Forest	Machine Learning	Fast, effective for high-dimensional data	Requires tuning, may over-flag rare lithologies	Unsupervised QC of full well log datasets
One-Class SVM	Machine Learning	Learns "normal" zones, robust to complex patterns	Sensitive to parameter tuning, slower on large data	Identifying geologically inconsistent values
Autoencoders	Deep Learning	Captures nonlinear patterns, good for reconstruction-based detection	Requires more data and computation	Large-scale denoising and unsupervised feature QC
Moving Average / Filters	Time- Series	Quick smoothing, highlights spikes and dips	Can smooth out real signals	Real-time QC or initial log smoothing
Multivariate Consistency Checks	Multivariate	Validates cross-log relationships (e.g., porosity from NPHI vs RHOB)	Needs calibrated rules or models	Advanced petrophysical interpretation workflows

Data Formats

Petrophysical data is not limited to numerical logs. A classification based on data format includes:

Continuous Numeric. Well log data – regularly sampled numerical (1D curves) like gamma ray, neutron porosity, bulk density, resistivity etc. (Xu et al., 2022), typically requiring normalization, smoothing, and imputation.

Categorical. Data that belongs to a distinct group, or categories based on shared characteristics or attributes such as petrophysical rock types, facies or fluid type etc.

Discrete. It represents limited numerical values such as number of fractures, bad hole flags, stratigraphic boundaries, or rock or fluid properties from core data.

Images or maps. Borehole image logs, thin sections and scanning electron microscope images, and 2D images generated from NMR logs etc. (Xu et al., 2022), processed using filtering, segmentation, and augmentation.

Textual/Unstructured. Includes mud log narratives, formation tops, or operational reports—analyzed using Natural Language Processing (NLP) methods.

Time-Series. Production and pressure datasets requiring feature extraction over time (e.g., moving averages, lags) and suited for sequence-based models like RNNs or LSTMs (Tachmazidis et al., 2022).

Selecting the right preprocessing and ML technique depends on both data format and analytical objective. Table 1 serves as a quick-reference framework, linking each data type to preprocessing steps and compatible ML models.

Petrophysical Data Analytics in ML

ML in petrophysics spans across four analytical categories:

Descriptive Analytics. Aims to summarize what has happened. Techniques include univariate statistics, correlation plots, and data profiling. These analyses validate data integrity and highlight patterns prior to model development (Bhattacharya, 2021).

Diagnostic Analytics. Explains why something occurred, using correlation matrices, multivariate plots, and Bayesian inference. For instance, unusual porosity-resistivity relationships may signal hydrocarbon presence or clay effects.

Predictive Analytics. Answers what is likely to happen. Using supervised learning methods like Random Forests, SVMs, and neural networks, models estimate unknown properties (e.g., permeability from wireline logs) in data-sparse intervals (Mohaghegh, 2017).

Prescriptive Analytics. Recommends optimal decisions. By integrating predictions with optimization algorithms, petrophysicists can refine well placement or data acquisition strategies (Alaudah et al., 2019).

Understanding these analytics categories helps petrophysicists to select appropriate ML models and define realistic expectations from each model class.

General ML Workflow for Petrophysical Solutions

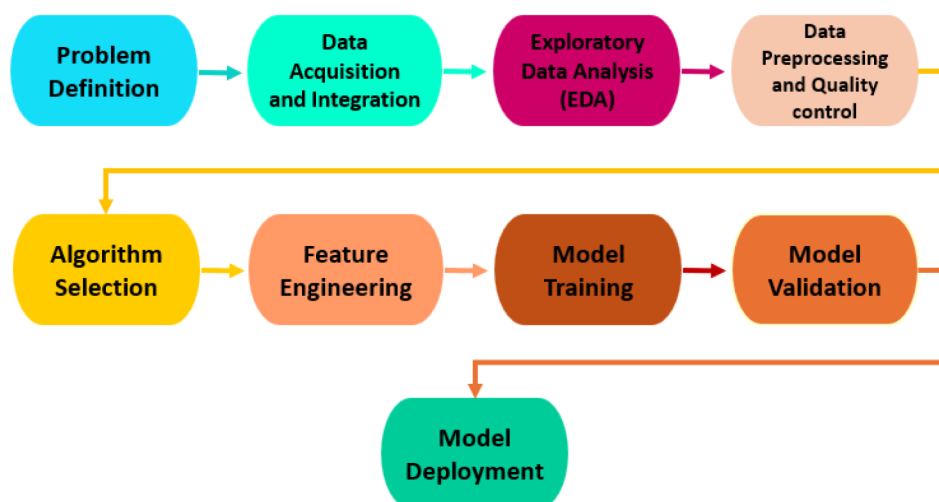


Figure 2—General ML Workflow for Petrophysical Solutions

Data Processing and Quality Control

Data quality is the single-most important factor in machine-learning projects, irrespective of the industry. The performance and reliability of any machine-learning model will depend on the quality of input data and if the input data used for doing the training is incomplete/ inconsistent, then the model learns wrong patterns based on this and predictions would be inaccurate. Poor quality data can introduce biases and ensuring good quality means we need to be aware about representativeness- this will also help the models generalize well to unseen data without overfitting (where the model is prone to picking up noise) or underfitting (where real patterns are missed). Clean and representative data reduces the need for complex models and also reduces the training time/ computational resources. One of the best papers reviewing various data quality considerations before doing petrophysical machine learning models was by [McDonald, Andrew \(2021\)](#) and he talked about the 6 data quality dimensions, as shown in the referenced [figure 3](#).



Figure 3—Data Quality Issues: Question Ask (After [McDonald Andrew, 2021](#))

Petrophysical logs are essential for formation evaluation, however before being used for interpretation, they generally need processing to account for operational issues, tool limitations and environmental factors. Modern datasets acquired with modern tools definitely have this issue to a lesser extent, but in case of older datasets (where data traceability and processing already applied also becomes a concern), these issues are more pervasive.

There are multiple issues, which impact petrophysical logs.

Noise on logs

We need to apply some de-noising or noise removal techniques

Depth-Matching

One of the most pervasive of issues, impacting data acquired from different tools and/ or in different runs in the borehole

Environmental Corrections

This one is a bit more nuanced topic because the main issue with environmental corrections is consistency and what we want is consistency in applying the corrections- it is difficult to tackle this purely by machine-learning

De-noising of Petrophysical Logs. Before machine-learning came into play, petrophysicists have been using various de-noising techniques like various statistical filtering methods (moving average, Fourier transforms etc.) but these have limitations.

- They might smoothen out real and important geological features
- They assume that the noise is also kind of uniform (does not change with depth) or that even noise has a certain kind of a structure.

These traditional filtering techniques might work in the case of background noise (as described in point b) but machine learning offers a more data-driven methodology to separate signal from noise without distorting real data, which carries proper geological information about heterogeneities.

The machine-learning approaches for denoising data on petrophysical logs can range from simplistic ones to more complicated ones and which one we should select definitely will vary from project to project.

Broadly, we can use

- Supervised Learning by training ML models on pairs of noisy and clean logs (if available) or synthetic data using models like Random Forest, Support Vector Regression (SVR), Gradient Boosting and Neural Networks.
- Unsupervised Learning using Autoencoders (Compressed representations of input logs and reconstruct denoised versions), Principal Component Analysis (PCA) (Identify and filter out noise by reducing dimensionality) and Self-Organizing Maps (SOM) (Cluster clean patterns and separate anomalies).
- Other advanced techniques like Signal Processing and Deep Learning

For all practical purposes, supervised learning models do a good job at denoising petrophysical logs.

Depth-Matching. Depth is a critical measurement for the successful development of an asset. For example, it allows the selection of perforation intervals and setting of packers at the correct depth within the well. All well log measurements are tied to depth. Ensuring that multiple logging measurements are at a consistent depth is essential to any petrophysical interpretation, reservoir analysis and multi-well correlations. (McDonald, Andrew, 2021) If data is not at a consistent depth when building machine learning

models, the algorithm will use incorrect data to build relationships with the target variable resulting in inaccurate predictions.

There have been multiple publications over the years proposing semi-automated and automated techniques to do depth-matching and surprisingly, this "simple" task has proved to be quite challenging because of multiple challenges with well log data e.g.:

- Varying SNR between different logging passes/ runs can cause changes in the shape of the log curve and this confuses any pattern recognition algorithm
- Differences in log measurement resolutions (sampling rates), also impacted by logging speeds- this can cause measurements to appear sparser, leading to missing some key peaks/ troughs, which are generally used for depth alignment

Whichever technique is used for depth-matching, it is a mandatory exercise, before applying any machine learning. Data should be depth shifted as accurately as possible prior to any multi-log processing/ interpretation using a single reference curve, and a record of these shifts should be documented in an audit trail

There have been some publications ([Liang et al, 2025](#)), talks about a non-machine-learning based multi-scale cross-correlation method while [Bhattacharya et al, 2025](#) compared multiple supervised and unsupervised machine learning models for well log data alignment) in recent times about automating depth-matching and these have shown promising results but these need to be evaluated more for more widely varying situations and datasets

Outliers. Outliers are abnormal data points that fall outside the expected range of values within a dataset. In well logging operations, these anomalies often result from borehole conditions such as washouts, sensor malfunctions, tool calibration issues, or rare geological formations. They may also stem from data acquisition errors or poor recording practices.

Identifying and handling outliers is a critical step before applying machine learning models. If not addressed, outliers can negatively impact model performance, leading to skewed results and unreliable predictions. This is especially important for algorithms like linear regression, which are highly sensitive to extreme values and can produce misleading outcomes when trained on noisy data.

A robust outlier detection and mitigation process improves data quality, enhances model accuracy, and ensures more reliable subsurface interpretations.

While all the outlier detection techniques need to be properly QC-ed by a properly trained petrophysicist (ideally experienced with the kind of formations being looked at, by these datasets), the visual and multi-variate techniques assume prior knowledge. The other techniques can be applied easier, but the limitations need to be kept in mind, properly.

Feature Engineering

Feature engineering is the process of selecting, transforming, and creating new variables from raw data to enhance the performance of predictive models.

In the context of machine learning applied to petrophysics, feature engineering can be divided into four main categories:

Feature Selection

Choosing the appropriate measurements to be input into the model.

Feature Creation

Deriving features such as porosity, saturation, and other computed properties from raw measurements for reservoir characterization.

Feature Transformation

Applying data science techniques such as normalization, standardization, and other augmentation methods to improve model performance.

Feature Selection. Feature selection involves identifying the most relevant measurements—such as logging data and core samples—to be used in the machine learning model. It is crucial to select input data that maintain sensitivity to the target variable being predicted. A balance must be struck by providing the model with sufficient informative features and avoiding the inclusion of irrelevant or insensitive data that could mislead probabilistic or clustering algorithms and degrade model performance.

Feature Creation. Feature creation involves computing new attributes from raw well measurements to enhance model robustness. For instance, derived properties such as porosity or water saturation can provide more interpretable and physically meaningful inputs. However, caution is advised: if a raw measurement is already being used directly in the model, reusing it to derive additional features may increase the model's dependency on that single measurement, potentially skewing the results.

Feature Transformation. Feature transformation techniques improve the performance of machine learning models by enhancing the statistical relationships between variables. Common methods applied to petrophysical logs include:

Standardization. Standardization adjusts the scale of log measurements to improve comparability across wells and measurements. This is typically achieved through rescaling techniques, such as feature rescaling, which normalize the range between the dataset's minimum and maximum values (Juszczak et al., 2002).

Normalization. Normalization is frequently applied when building machine learning models for petrophysical data. "The idea behind data normalization is to diminish or eliminate difference among data sets that are not related to geological differences or reservoir properties such as the effect that bad borehole or wrong calibration can caused on the data, it also allows to compare similar measurements coming from different vendors or old vs new technology (The process of data normalization is the adjustment to values between 0 and 1, derived from the minimum and maximum values, respectively" (Vega-Ortiz et al., 2020)

Other Augmentation Techniques: Additional data augmentation techniques can further enhance model reliability and include:

- Principal Component Analysis (PCA)
- Singular Value Decomposition (SVD)
- Discrete Wavelet Transform (DWT)

These methods reduce dimensionality, capture essential patterns, and remove noise, helping to generate more informative features for training.

Digital Ready Data Using Advanced Petrophysical Techniques. While traditional transformation techniques rely on general data science methods, advanced petrophysical approaches can also be employed to generate "Digital Ready Data." A notable example is High-Angle, Horizontal well-log interpretation model-compare-update methodology.

"Complex geometry effects observed on log responses in high-angle and horizontal wells can create challenges in achieving accurate formation evaluation. The situation is made even more difficult in thinly bedded reservoirs where measurements may respond to multiple layers within their volume of investigation" (Salehi et al., 2020).

High-Angle, Horizontal well log interpretation model-compare-update methodology in petrophysics enables semi-automated construction of local layer geometries and properties using LWD density images. Once the geometry and layer positions are defined, inversion methods are applied to extract true layer

properties, corrected for geometrical effects. These geometrically corrected logs serve as high-quality inputs for machine learning models, enhancing model robustness, particularly for techniques based on clustering or probabilistic inference.

Machine Learning Algorithms for Petrophysical Applications

Machine learning algorithms play an increasingly central role in petrophysical analysis. Supervised learning techniques, particularly regression and classification, are widely used to predict either categorical properties e.g. facies, fracture, faults etc. or continuous reservoir properties e.g. Porosity, permeability, compressional slowness etc.

Figure 4 provides a classification of machine learning approaches used in petrophysics, grouped by learning type and algorithmic structure. Table 3 complements this figure by mapping these approaches to real-world petrophysical use cases, helping practitioners make informed decisions during model selection, data preparation, and validation.

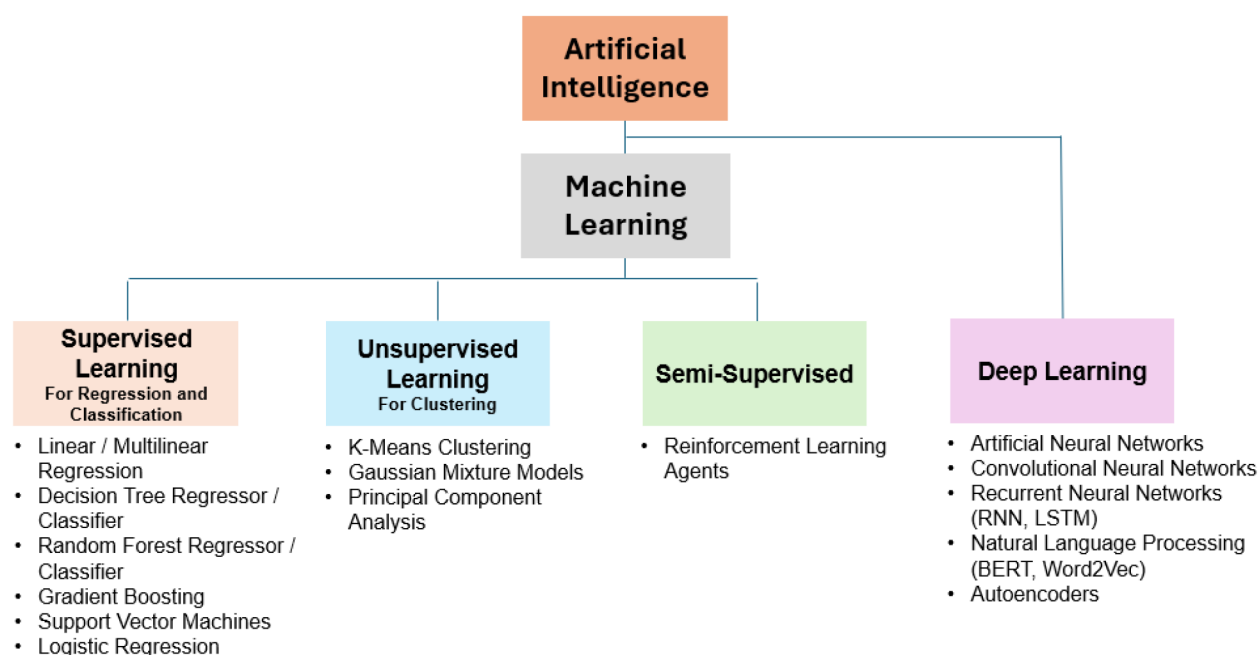


Figure 4—Classification of Machine Learning Algorithms Applied in Petrophysics: Hierarchical classification of artificial intelligence techniques, emphasizing machine learning subtypes—supervised, unsupervised, semi-supervised, and deep learning—with associated algorithms commonly used in petrophysical workflows

Before model selection, it is essential to conduct Exploratory Data Analysis (EDA) to understand the structure, quality, and distribution of the data. EDA helps identify trends, outliers, missing values, and feature correlations. It also assists in selecting the most relevant input variables and understanding the relationships between them—both of which are critical for building effective machine learning models.

Every machine learning project starts with choosing the right algorithm and selection of algorithm should reflect both the structure of the available data (labeled, unlabeled, or mixed) and the geological context of the problem. In this paper, we are discussing only the popular supervised and unsupervised Machine Algorithm. SSL and DL is beyond the scope of this paper, however for completeness of discussion SSL and DL subjects may appear.

Table 3—Classification of Machine Learning Approaches with Petrophysical Use Cases:
Comparison of machine learning approaches based on input data requirements, problem type, algorithmic methodology, and representative applications in petrophysical workflows.

Learning Type	Input Data Type	Type of Problem Solves	How it works	Petrophysical Use Cases
Regression	Labeled continuous log data (e.g., GR, density, resistivity)	Predicting continuous properties like porosity or permeability	Learns mapping between inputs and a continuous target using error minimization	Estimating permeability from well logs using Random Forest or XGBoost in data-sparse zones (Zhang et al., 2019)
Classification	Labeled categorical log data (e.g., lithofacies, rock type)	Classifying facies, lithology, or reservoir	Learns decision boundaries between predefined classes	Lithofacies classification using gamma ray, resistivity, and neutron logs (Alaudah et al., 2019)
Unsupervised	Unlabeled log data (e.g., GR, resistivity)	Identifying electrofacies or rock types without	Clusters similar data points based on feature similarity	Electrofacies identification in old wells without core or facies labels using K-Means clustering (Mohaghegh, 2017)
Semi-Supervised	Partially labeled logs (e.g., labeled core intervals + unlabeled logs)	Classifying data when labeled examples are limited	Combines labeled and unlabeled data to improve learning	Propagating limited core-labeled facies into surrounding well intervals via graph-based learning (Alaudah et al., 2019)
Deep Learning	Structured, unstructured, or image data (e.g., logs, text, core images)	Complex tasks like image interpretation or production forecasting	Uses layered neural networks to automatically extract complex features	Core image-based pore type classification and borehole feature detection using CNNs (Karimpouli and Tahmasebi, 2020)

Linear and Multi-linear Regression

Linear and multiple linear regression (MLR) are foundational supervised learning algorithms used to predict continuous reservoir properties such as porosity and permeability. Their primary advantages lie in interpretability, simplicity, and computational efficiency, making them suitable for early-stage modeling and geological validation.

Linear regression models the relationship between a single predictor and the target using a straight line. MLR extends this concept to multiple predictors, fitting a regression plane (in three dimensions) or a hyperplane (in higher dimensions) that best explains the target variable based on available inputs (Zhang et al., 2019).

These models estimate coefficients by minimizing the error between predicted and observed values. Figures 5a and 5b illustrate these concepts: Figure 5a shows a single-variable regression using density logs, while Figure 5b presents a multivariate regression surface using both density and compressional slowness.

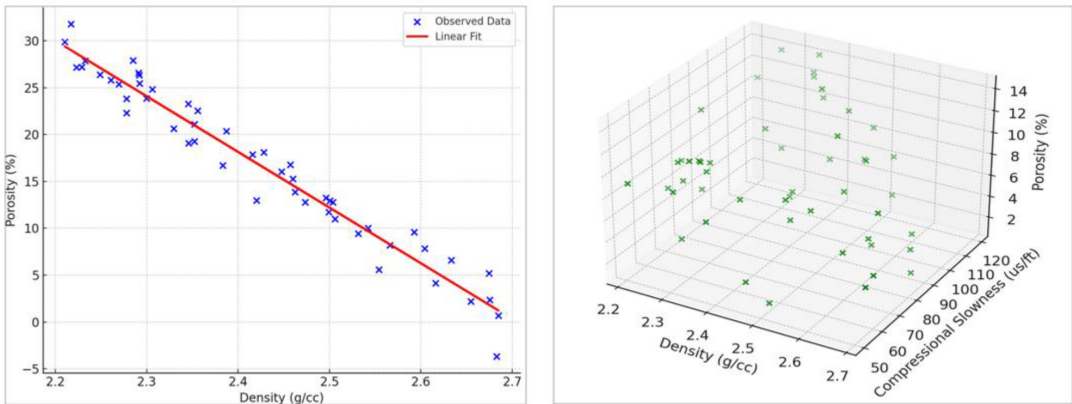


Figure 5—The core concept of linear and multiple linear regression for petrophysical applications:
Figure 5a (Left): Linear Regression: Shows porosity estimation based on a linear relationship with density log data. Figure 5b (Right): Multiple Linear Regression: Illustrates porosity estimation using both density and compressional slowness (Dt) as predictors, visualized in 3D.

MLR assumes linear relationships between predictors and targets, low multicollinearity among inputs, independence of errors, and constant variance (homoscedasticity). These assumptions often hold in clean, homogeneous formations such as sandstones. However, in geologically complex settings (e.g., carbonates), nonlinearities and variable coupling may limit the effectiveness of MLR.

One of the practical benefits of MLR is its ability to extrapolate reasonably well beyond the range of the training data—something many nonlinear models (e.g., decision trees, random forests) struggle with. This is because the linear nature of the model extends the learned relationships indefinitely in the same direction, making it more robust for predictions in under-sampled or unseen input ranges. However, care must still be taken to ensure that extrapolated results are geologically reasonable.

Exploratory Data Analysis (EDA) plays a key role in diagnosing violations of these assumptions and in selecting relevant features. EDA also helps identify important input variables and underlying data correlations, which directly influence model reliability. Feature engineering—such as variable transformation—may further enhance model performance.

Decision Trees

Decision Trees is simple, interpretable models used for both regression and classification that split data into hierarchical rules based on input features like gamma ray or density. They do not require assumptions about linearity, normality, or feature independence, making them suitable for complex but rule-driven petrophysical patterns. Figure 6 shows how the model conceptually generates a stepwise prediction reflecting such transitions.

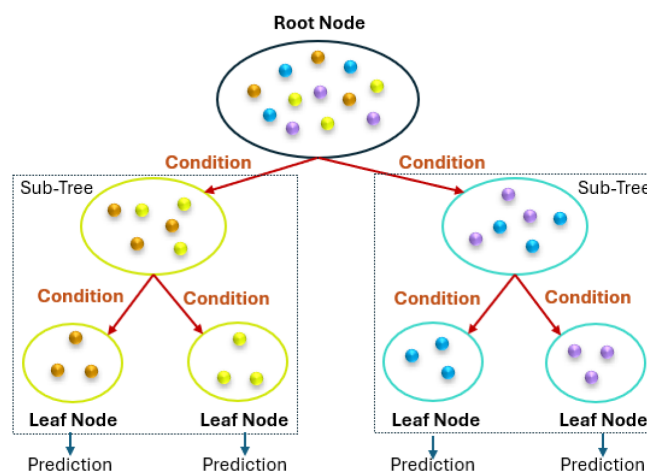


Figure 6—A decision tree structure illustrating the concept.

Decision Tree Regression predicts continuous properties such as porosity or permeability by recursively splitting data to reduce variance. It performs well in formations with abrupt property changes—e.g., stratigraphic boundaries or laminated zones. On the other hand, Decision Tree Classification assigns categorical labels (e.g., lithofacies, fluid zones) using sequential feature-based splits to maximize class homogeneity. The method is favored for its explainability and fast computation, especially in early-stage workflows (Alaudah et al., 2019; Geron, 2019).

In complex geological environment, decision trees are occasionally used in petrophysical analysis due to their interpretability, they are generally outperformed by ensemble methods like Random Forest and Gradient Boosting – offer greater accuracy and robustness.

Although easy to interpret, trees can overfit if allowed to grow too deep, especially with noisy data. Regularization (e.g., limiting depth or requiring minimum samples per split) improves generalization (Gul et al., 2020).

Random Forest Algorithms for Regression and Classification

Random Forest is an effective ensemble-based supervised learning algorithm that has shown strong performance across diverse petrophysical applications – used for both regression and classification, depending on whether the target is continuous or categorical. It often outperforms linear models in complex formations where input–target relationships are nontrivial (Zhang et al., 2019). This technique outperforms single decision tree as it builds multiple decision trees, which means it tends to reduce overfitting and manage noisy data more effectively (Breiman, 2001; Zhang et al., 2019; Alaudah et al., 2019). Eventually, it outputs more accurate predictions compared to linear models.

Random Forest Regression is effective for predicting continuous reservoir properties, especially when relationships with well logs are nonlinear or noisy. It shows a smoother and more generalizable regression output compared to a single tree. On the other hand, Random Forest Classification is widely used for subsurface classification tasks like lithofacies and electrofacies prediction. It assigns class labels by majority voting across trees.

A key advantage of Random Forest is, during training, it selects the features randomly (bootstrapping sampling method) that increases the diversity among the trees – improves overall performance. It works well with high-dimensional input data; however, tree depth should be tuned carefully—especially with limited data—to prevent overfitting. The disadvantage of this method is, it generally cannot extrapolate beyond the training range and may become computationally expensive for large datasets or many trees. A large amount of input features increases the potential for deeper trees, but feature sub-sampling (e.g., `max_features`) helps control this complexity.

Gradient Boosting (XGBoost, LightGBM)

Gradient Boosting algorithms—such as XGBoost and LightGBM—are widely used in petrophysical modeling for their ability to capture complex, nonlinear relationships in well log data. These methods are especially effective in heterogeneous formations where linear models or single decision trees fail to generalize.

Unlike Random Forest, Gradient Boosting builds trees sequentially, with each new tree trained to correct the residuals of the previous ensemble. This stage-wise learning optimizes a loss function (e.g., mean squared error), leading to highly accurate models over successive iterations. Figure 7 compares a single shallow decision tree with a gradient boosting ensemble, which yields a smoother and more refined prediction of reservoir properties.

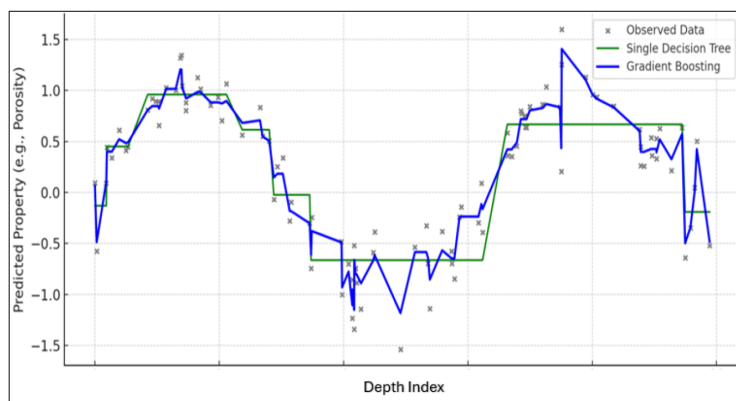


Figure 7—Gradient Boosting vs Single Decision Tree: Conceptual comparison of porosity prediction using a single shallow tree and a Gradient Boosting ensemble. Boosting improves prediction accuracy by iteratively reducing residual errors through sequential tree refinement.

Gradient Boosting does not assume linearity or specific data distributions, and it is robust to varying feature scales, eliminating the need for standardization. It effectively models high-dimensional interactions

and performs well in datasets with subtle lithological variations or high noise levels. However, careful tuning of hyperparameters—particularly tree depth, learning rate, and number of trees—is essential to avoid overfitting and ensure generalization. The flip side of this technique is, it is computationally more intensive than simpler models and may lack full transparency (black-box model).

In petrophysical applications, Gradient Boosting is commonly used for predicting continuous properties such as porosity, permeability, and water saturation. For instance, [Zhang et al. \(2019\)](#) showed that XGBoost outperformed traditional regression methods in permeability prediction from wireline logs in unconventional reservoirs.

Support Vector Machine (SVM)

Support Vector Machines (SVMs) are supervised learning algorithms commonly used for classification tasks, especially when class boundaries are complex and nonlinear. In petrophysics, SVM is effective in scenarios where simpler models fail to separate classes such as lithofacies or fluid types. It works by identifying an optimal hyperplane that maximizes the margin between classes, as shown in [figure 8a](#) ([Bhattacharya, 2021](#)).

Support Vector Machines (SVMs) are occasionally applied in petrophysical classification tasks, particularly when data is limited and classes are well separated. However, due to scalability issues, lack of interpretability, and sensitivity to feature scaling, SVMs are generally less favored than tree-based or neural network models in modern petrophysical workflows.

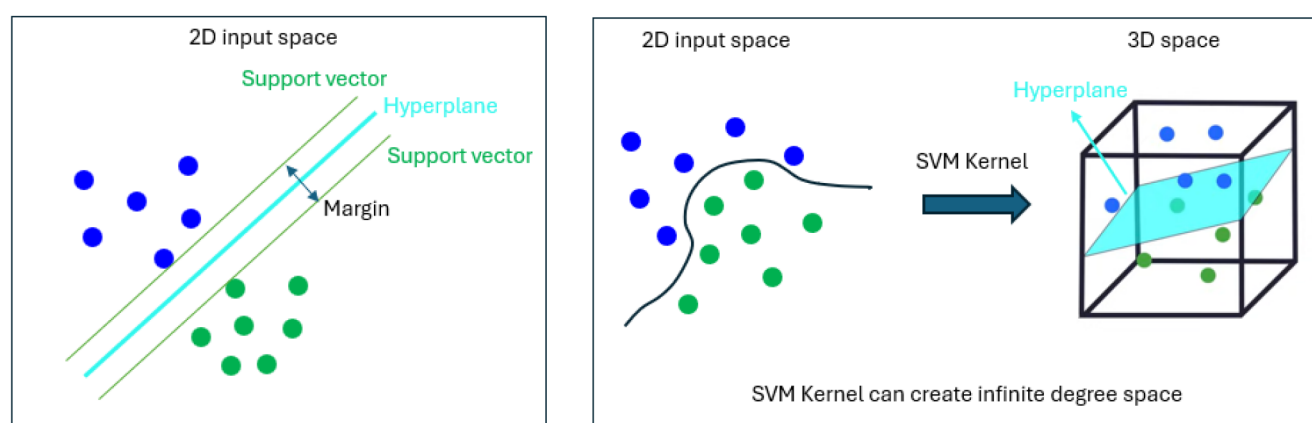


Figure 8—Support Vector Machine (SVM) classification: Figure 7a (left): Illustrating when the classes are linearly separable uses support vectors to optimize the margin. Figure 7b (right): when classes are non-linearly separable uses SVM Kernel to transform into higher-dimensional.

When data is not linearly separable, SVM uses kernel functions to project input features into a higher-dimensional space where separation becomes possible. [Figure 8b](#) illustrates how this transformation enhances class separability. Kernel choice and regularization are critical to performance.

Although originally designed for binary classification, SVMs can be extended to multi-class problems via one-vs-one or one-vs-rest strategies. They have been applied to vuggy porosity prediction using log, NMR, and core data ([Howat et al., 2016](#); [Deng et al., 2019](#)), as well as permeability and facies classification ([Zhang et al., 2019](#)).

SVMs are most useful in high-dimensional, low-sample-size problems. However, they are computationally intensive on large datasets, sensitive to feature scaling, and offer limited interpretability compared to decision trees or Random Forests.

Logistic Regression

Logistic regression is a simple, interpretable classification algorithm used for binary and multi-class outcomes derived from continuous or categorical predictors. In petrophysics, it is often applied to classify facies or fluid types due to its transparency and computational efficiency (Gouda and Salim, 2020).

The model estimates the probability that an input belongs to a specific class using a sigmoid function, then assigns a class based on a decision threshold. Figure 9 illustrates a sand–shale classification example, where points are assigned based on whether their predicted sand probability exceeds the threshold. In the same study, fluid classification was performed using 14 elastic attributes to estimate gas-bearing zones (Gauda and Salim, 2020).

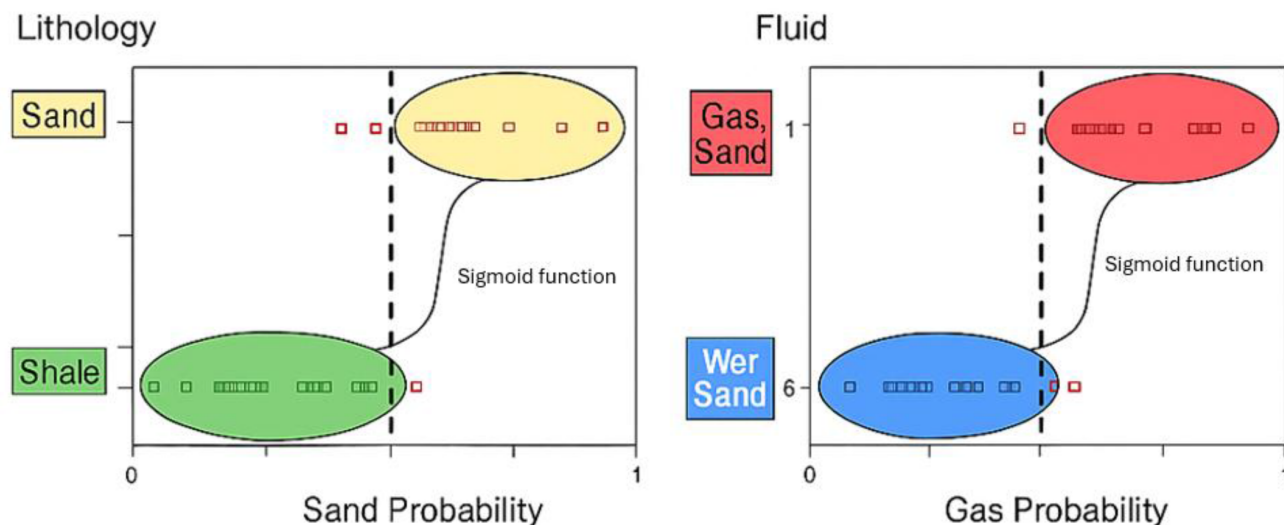


Figure 9—Logistic Regression Lithofacies Classification Using Sand / Gas Probability: This figure illustrates the classification of lithology (sand vs. shale) and fluid (gas vs. wet sand) using a logistic regression model. The sand probability predictor is mapped to a sigmoid curve, and a threshold is applied to distinguish between classes. Data points above the threshold are classified as sand / gas, while those below are identified as shale / wet sand (Modified after Gouda and Salim, 2020).

Logistic regression assumes a linear relationship between input features and the log-odds of the target class. It works well when class boundaries are linear, and the features are not strongly correlated. However, it performs poorly with overlapping classes, non-linear patterns, or imbalanced data—unless extended with feature transformations or regularization.

While Logistic Regression is still used as a baseline model for its interpretability, it is often outperformed by more flexible methods like Random Forest and Gradient Boosting in complex geological settings.

K-Means Clustering

K-Means is a widely used unsupervised learning algorithm that identifies natural groupings in data without requiring labeled outputs. In petrophysics, it is particularly useful when no core or labeled facies data are available. It enables clustering of depth-indexed log responses based on their similarity, even in complex or transitional geological settings (Bhattacharya, 2021).

The algorithm partitions data into a predefined number of clusters (k) by minimizing within-cluster variance. It begins by randomly initializing k centroids in feature space (e.g., gamma ray vs. density). Each data point is assigned to the nearest centroid using Euclidean distance, forming initial clusters. The centroids are then recalculated as the mean of the assigned points, and the process repeats until convergence, or a maximum number of iterations is reached (Figure 10).

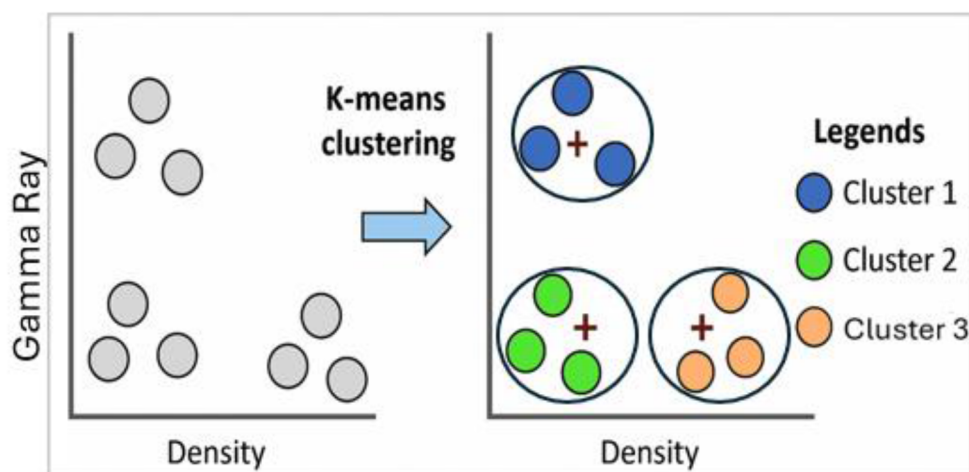


Figure 10—K-Means Clustering Process for Electrofacies Classification: The diagram illustrates iterative centroid movement and cluster assignment using gamma ray and density features. Data points are grouped into $k = 3$ clusters, with centroids updated after each iteration. The process continues until stable cluster boundaries emerge, useful in distinguishing facies without core labels (Modified after [Bhattacharya, 2021](#)). K-Means assumes that clusters are spherical, equally sized, and linearly separable in Euclidean space. It also requires the number of clusters (k) to be known in advance, which can be limiting in heterogeneous formations.

In petrophysical applications, K-Means has been used for electrofacies classification in carbonate reservoirs using gamma ray, neutron, and density logs. Without core data, it identified log-based clusters corresponding to depositional facies, aiding zonation and reservoir property modeling ([Bhattacharya, 2021](#)).

However, K-Means is sensitive to the initial centroid positions and performs poorly with non-spherical or overlapping clusters—common in depositional environments influenced by diagenesis. Preprocessing (e.g., feature scaling) and domain knowledge are often needed to improve clustering quality.

Model evaluation typically involves silhouette scores, inertia (within-cluster sum of squares), or comparison with core-based or mud log interpretations. Visual tools such as cross-plots and depth tracks remain essential for validating the geological relevance of the clusters.

Algorithm Selection Guide

A structured and thoughtful approach to algorithm selection is critical for effective machine learning applications in petrophysics. A quick tip for algorithm selection is given below.

- First start with interpretable models such as linear regression and decision trees, as these offer clear, early-stage insights into data behavior
- Transition to ensemble techniques such as Random Forests or XGBoost When predictive accuracy becomes essential or the data exhibits complex, nonlinear relationships.
- Use Support Vector Machines (SVM) for specialized scenarios involving small datasets with high complexity, which provide robust performance and decision-making precision.
- K-Means or Gaussian Mixture Models (GMM) a clustering algorithm, choose in situations where labeled data is unavailable, and the goal is exploratory analysis – very effective for identifying hidden structure or grouping lithofacies.

Regardless of the algorithm chosen, it is imperative to align its underlying assumptions with geological context—cross plots, correlation analysis, and domain knowledge should always inform the selection process to ensure the model's physical plausibility and relevance to subsurface interpretation.

[Table 4](#) illustrates strength, limitations, and recommended uses of all the algorithms discussed in the previous section.

Table 4—A quick reference guide for ML algorithms selection

Algorithm	Strengths	Limitations	Recommended Use Context
Linear & Multiple Linear Regression	Interpretable; fast computation; detects linear trends	Assumes linearity; sensitive to multicollinearity and outliers	Use when relationships are assumed linear or as a baseline model
Decision Tree (Regression /- Classification)	Easy to visualize; handles mixed data types	Prone to overfitting; unstable with small changes in data	Use for transparent rule-based models or quick interpretability
Random Forest (Regression /- Classification)	Robust to overfitting; ranks feature importance; handles missing values	Slower for large datasets; less interpretable than single trees	Use for nonlinear tasks, feature selection, and improved generalization
Gradient Boosting (e.g., XGBoost)	High accuracy; handles complex feature interactions; robust to missing data	Computationally intensive; needs careful tuning	Use when high prediction accuracy is required, with extensive tabular features
Support Vector Machines (SVM)	Effective in high-dimensional space; works well with small datasets	Difficult to scale to large datasets; kernel selection can be complex	Use when data is small and boundaries between classes are nonlinear
Logistic Regression	Probabilistic output; interpretable; simple baseline for binary classification	Assumes linear decision boundary; limited expressiveness	Use for binary classification where interpretability and probability scores are useful
K-Means Clustering	Fast; simple; effective for exploratory analysis	Requires predefined number of clusters; sensitive to scaling	Use for unsupervised rock typing or lithofacies grouping
Gaussian Mixture Models (GMM)	Probabilistic clustering; flexible with overlapping distributions	Sensitive to outliers; assumes Gaussian distribution	Use for facies clustering when classes may overlap, and soft membership is valuable

Practical Considerations and Challenges

The application of machine learning to petrophysical data offers significant potential for automating interpretation, improving reservoir characterization, and supporting decision-making in complex geological environments. However, the performance and reliability of these models are highly dependent on several practical considerations. These include not only technical aspects such as data partitioning, class balance, and metric selection, but also domain-specific issues such as interpretability, spatial bias, and geological plausibility. This chapter outlines six key considerations essential for developing robust, geologically meaningful models in petrophysics.

Data Splitting and Model Validation

Petrophysical datasets often exhibit strong spatial continuity, especially within a single wellbore. As a result, random row-wise splitting of data can introduce significant data leakage, whereby patterns learned from training data are inadvertently reflected in the test set. This can lead to overestimated performance metrics that do not generalize to new wells or formations.

To mitigate this issue, well-based data splitting is recommended. In this approach, entire wells are assigned exclusively to either the training or testing sets. This simulates real-world scenarios where the model must be deployed on unseen wells.

Class Imbalance

Imbalanced class distributions are common in petrophysical classification tasks. For example, shale or water-bearing intervals may dominate the dataset, while hydrocarbon-bearing sands or rare facies are underrepresented. This imbalance can bias the model toward majority classes and result in poor performance on the classes of greatest interest.

Several strategies are available to address class imbalance. These include applying class weights during model training, oversampling minority classes, and under sampling dominant classes. It is also important

to use evaluation metrics that reflect class-wise performance, as overall accuracy can be misleading in imbalanced scenarios.

Performance Metrics

Selecting appropriate performance metrics is essential for accurately evaluating model quality. In regression tasks, the root mean squared error (RMSE) is widely used because it penalizes large deviations more heavily, which is important when extreme errors are geologically significant. The mean absolute error (MAE) offers a more interpretable average error and is less sensitive to outliers. The coefficient of determination (R^2) provides insight into the proportion of target variance explained by the model. Cross-plots of predicted versus measured values are also a standard diagnostic in petrophysical workflows, helping to visually assess trend fit, bias, and residual spread.

For classification tasks, the F1 score—particularly its macro or weighted versions—is preferred in imbalanced settings, as it balances precision and recall across all classes. While accuracy is simple to understand, it can be misleading when class distributions are skewed. The confusion matrix offers a detailed breakdown of correct and incorrect predictions by class. For binary classification problems, the area under the receiver operating characteristic curve (ROC-AUC) provides a threshold-independent measure of class separability. Cross-validation is essential for confirming the generalizability of any chosen metric across multiple wells or intervals.

Model Interpretability

Model interpretability is a critical consideration in petrophysical workflows, where predictions must often be justified to domain experts. While complex models such as ensemble methods and neural networks may offer high accuracy, they often lack transparency. This can hinder their acceptance in environments where decision-making requires explainable logic.

To address this, interpretable models such as logistic regression or decision trees are often used as baselines. When using more complex models, interpretability tools such as SHAP (SHapley Additive exPlanations), permutation importance, and LIME can be employed to highlight which features contribute most to predictions. These tools help bridge the gap between predictive performance and geological reasoning.

Temporal or Spatial Bias

A common but underappreciated issue in petrophysical modeling is temporal or spatial bias. Models trained on a narrow set of wells from a single field, formation, or campaign may fail to generalize to different regions or drilling periods. This can lead to poor performance when the model is applied in broader operational contexts.

To mitigate spatial and temporal bias, models should be trained and validated across wells from diverse geological settings, formations, or time periods. This ensures that the model learns generalizable patterns rather than local correlations. Explicit spatial or stratigraphic validation can further quantify how well a model generalizes beyond its training domain.

Geological Plausibility and Post-Processing

Statistical performance alone does not guarantee that machine learning predictions are geologically reasonable. A model may achieve high accuracy while producing outputs that violate known depositional sequences, structural boundaries, or log-based interpretations. Therefore, geological plausibility must be incorporated into the validation and refinement process.

Manual QC by experienced petrophysicists remains an essential step in validating the final outputs. Ensuring geological plausibility not only improves model trust but also enhances its usability in integrated workflows such as geomodeling and reservoir simulation.

Future Trends

Moving from purely data-driven to physics-informed ML

As machine learning advances in petrophysics, Physics-Informed Machine Learning (PIML) emerges as a transformative approach by embedding geological and physical constraints directly into learning algorithms. Unlike purely data-driven models, PIML integrates laws such as Darcy's law or wave propagation equations to enhance generalization, reduce non-physical predictions, and improve reliability under data-scarce conditions. Future applications may include real-time reservoir simulation, facies classification, and multi-scale data fusion (Raissi et al., 2019; Moseley et al., 2019; Bhattacharya, 2021), making PIML a promising direction for robust and physically consistent subsurface modeling.

Cross Domain Models

Cross-domain modelling is poised to transform petrophysics by integrating seismic, well log, and core data into unified machine learning frameworks, enabling multi-scale, geologically consistent predictions. This integration enhances interpretability and reduces uncertainty in subsurface characterization. Emerging techniques such as ensemble learning and multi-resolution neural networks will enable petrophysicists to model complex reservoirs with greater accuracy, even in data-sparse settings—reshaping workflows from static interpretation to dynamic, data-driven reservoir management (Zhang et al., 2019; Mohaghegh, 2017; Dubrule, 2017).

Model training in continuous learning System

Continuous learning systems are redefining model deployment in petrophysics by enabling ML algorithms to adapt dynamically to new data streams from drilling, production, and surveillance. Unlike static models, these systems evolve over time, improving prediction accuracy and operational decision-making. Techniques such as online learning and transfer learning offer scalable solutions for real-time model refinement, making them vital for future digital oilfield strategies (Bhattacharya, 2021; Mohaghegh, 2017; Tachmazidis et al., 2022).

Real-time analytics with Edge Computing

Edge computing enables real-time analytics directly at the data source—on rigs or wellsites—minimizing latency and bandwidth limitations associated with cloud processing. In petrophysics, this advancement allows immediate interpretation of LWD, mud logs, and production data, enhancing decision-making during drilling and formation evaluation. Combined with ML, edge platforms empower adaptive workflows and early anomaly detection, accelerating automation in digital oilfields (Bhattacharya, 2021; Khoshnood et al., 2020; Mahmoud et al., 2021).

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